

pigeonhole principle $\Rightarrow$ ^{pigeons} 6 P_1 P_2 P_3 P_4 P_5

If n pigeonholes are occupied by $n+1$ or more pigeons then at least one pigeonhole is occupied by more than one pigeon.

Example $\Rightarrow$ If a class contains 13 students then two of the students (pigeons) have been in the same month (pigeonholes).

Generalized pigeonhole principle $\therefore$

If n pigeonholes are occupied by $kn+1$ or more pigeons, where k is a positive integer, then at least one pigeonhole is occupied by $k+1$ or more pigeons.

Find the minimum no. of students in a class to be sure that three of them are born in the same month.

$$n = 12 \text{ (pigeonholes)} \quad k+1 = 3 \Rightarrow \boxed{k=2}$$

$$\begin{aligned} \underline{kn+1} &= 2(12)+1 \\ &= 25 \text{ students} \end{aligned}$$

Minimum

Suppose a bag contains many red, white and blue socks. Find the minimum no. of socks that one needs to choose in order to get two pairs (four socks) of the same color.

$$n = 3$$

$$\begin{aligned} \underline{k+1} &= 4 \\ \Rightarrow k &= 3 \end{aligned}$$

$$n=3$$

$$\Rightarrow k=3$$

$$k_{n+1} = 3(3) + 1 = 10$$

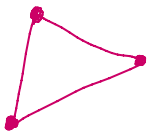
Find the minimum no. of students needed to guarantee that
five of them belongs to the same class (BIA I, MCA I, MCA II, BCA II)

$$\rightarrow n=4$$

$$\rightarrow k+1=5 \Rightarrow k=4$$

$$k_{n+1} = 4(4) + 1 = 17$$

Three persons enter a railway compartment. If there are 5 seats
 vacant. In how many ways they take these seats



$${}^5P_3 = \frac{5!}{(5-3)!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{2!} = 60$$

The number of triangles that are formed by choosing the vertices from a set of
12 points, seven of which lie on the same line is

(A) 105

(B) 15

(C) 175

(D) 185

$${}^{12}C_3 - {}^7C_3 = \frac{12!}{3!(9)!} - \frac{7!}{3!4!} = \frac{12 \times 11 \times 10 \times 9!}{3 \times 2 \times 1 \times 9!} - \frac{7 \times 6 \times 5 \times 4!}{6 \times 4!} = 220 - 35$$

The total number of 9 digit numbers which have all different digits is

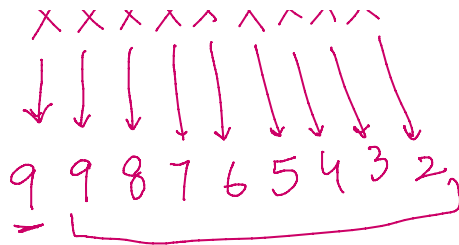
(A) 10!

(B) 9!

(C) 9 × 9!

(D) 10 × 10!





$$\underline{9 \times 9!}$$

4. In how many ways can 10 examination papers be arranged so that the best and the worst papers never come together?

- ✓ ☒ A. $8 \times 9!$
☐ B. $8 \times 8!$
☐ C. $7 \times 9!$
☐ D. $9 \times 8!$

$$\underline{\text{Total}} = 10!$$

$$\text{Best and worst papers together} = 2 \times 9!$$

$$\begin{aligned} \text{Not together} &= 10! - 2 \times 9! \\ &= 10 \times 9! - 2 \times 9! = (10-2)9! \\ &= 8 \times 9! \end{aligned}$$

